

AdvNLP Exam Cheat Sheet 2 POS Tagging / HMM / Viterbi A4 / 4 pp

Classification Evaluation (L4 ★★)

Confusion matrix: rows = ground truth, cols = prediction; diagonal = correct.

Per class c (TP/FP/FN from that class):

$$P_c = \frac{TP}{TP+FP} \quad R_c = \frac{TP}{TP+FN}$$

$$F1_c = \frac{2P_cR_c}{P_c + R_c} \quad \text{Acc} = \frac{\sum_c TP_c}{N}$$

N/A cases: no predictions for $c \Rightarrow P=N/A$ (0/0); no true $c \Rightarrow R=N/A$; either undefined $\Rightarrow F1=N/A$.

Averaging:

$$\text{macro-}F1 = \frac{1}{C} \sum_c F1_c$$

$$\text{weighted-}F1 = \frac{\sum_c n_c F1_c}{\sum_c n_c} \quad (n_c = \text{support})$$

micro- $F1$: pool all TP/FP/FN then one $F1$ ($\approx \text{acc}$).

macro = each class equal (hurts on bad rare class); **weighted** = by frequency.

Recipe: build matrix $\rightarrow$ per-class $P, R, F1 \rightarrow$ macro=avg, weighted=avg by support. Watch 0/0.

Pitfall: accuracy can be 1.0 yet useless if test set misses classes.

Dependency Parsing + UAS/LAS (L5 ★★)

Arc-standard transition parser; config = (Stack, Buffer, Arcs). Stack top = B , 2nd-top = A :

- **SHIFT:** buffer front $\rightarrow$ stack.
- **LEFT-ARC(r):** arc $B \rightarrow A$ (B head of A), pop A .
- **RIGHT-ARC(r):** arc $A \rightarrow B$ (A head of B), pop B .

Repeat until buffer empty & stack=[ROOT].

UAS = $\frac{\# \text{correct heads}}{\# \text{tokens}}$ (label ignored)

LAS = $\frac{\# \text{correct head\&label}}{\# \text{tokens}}$

Always $\text{LAS} \leq \text{UAS}$. Right head + wrong label $\Rightarrow$ UAS yes, LAS no. (Root usually excluded.)

POS Tagging / HMM / Viterbi (L5)

Baselines: dict 85%, most-freq-tag 90%, SOTA 97%, human 98%.

HMM: states=tags(hidden), obs=words. Assumptions: transition $P(t_i|t_{i-1})$, emission $P(w_i|t_i)$. Tag via Bayes: $\hat{t} = \arg \max_t P(w|t)P(t)$.

Viterbi (DP, best path): $V_{i,t} = \max_{t'} V_{i-1,t'} P(t|t')P(w_i|t)$, cost $O(n|T|^2)$.

Assumptions: **Markov** (tag depends only on prev tag) + **output indep** (word depends only on its tag). Unseen word $\Rightarrow$ emission 0 (needs smoothing). **CRF:** globally normalized, models features over whole seq, less **label bias** than MEMM.

Preprocessing (L1 ★)

Tokens=all occurrences; **types**=distinct tokens. Ex: "The big dog ate my exam questions and the smaller dog ate my homework questions" $\Rightarrow$ **15 tokens, 10 types**.

Lemmatize (needs PoS): ate $\rightarrow$ eat, smaller $\rightarrow$ small, "better"(adj) $\rightarrow$ good. Stem=crude suffix chop.

Stopword removal drops function words. **Traps:** removing **negation** "not" flips sentiment to positive; lowercasing "May" $\rightarrow$ "may" loses month/proper-noun (hurts NER/PoS).

Naive Bayes + Smoothing (L4)

$\hat{c} = \arg \max_c P(c) \prod_i P(w_i|c)$, $P(w|c) = \frac{\text{count}(w,c)}{\sum_{w'} \text{count}(w',c)}$. Token only in class $A \Rightarrow P(w|A)>0$, $P(w|B)=0$; one zero kills whole product.

$P(w|c) = \frac{\text{count}(w,c)+\alpha}{\sum_{w'} \text{count}(w',c)+\alpha|V|}$
Laplace=additive with $\alpha=1$: prevents zeros without distorting nonzero probs. Use **log** probs (avoid underflow).

Word Embeddings (L1)

Distributional semantics: "know a word by the company it keeps" (Firth). **One-hot** limit: no similar-

ity between words, sparse, OOV-fail.

word2vec = shallow 2-layer NN.

Skip-gram: predict context from center word; logistic/sigmoid on $\text{dot}(\text{center}, \text{context}) \rightarrow \text{prob}$. small window=syntactic, large=semantic. **SGNS:** update few negatives ($\propto f^{0.75}$) not full $|V|$ softmax. **CBOW:** predict center from avg context.

GloVe=count-based, global co-occurrence matrix factorization (**not** predictive). **fastText**=sum of char n-grams $\Rightarrow$ handles OOV/typos/morphology. All three **static** (1 vec/word) $\Rightarrow$ fail polysemy ("bank"); BERT contextual fixes it. Analogy king-man+woman $\approx$ queen.

WSD via WordNet: avg embedding of each sense's gloss words, compare to sentence avg by **cosine**, pick max.

Syntax: Constituency & PCFG (L5)

Constituency=phrase tree (S $\rightarrow$ NP VP; NP, VP, PP); **dependency**=head $\rightarrow$ dependent arcs. Both are trees, different views.

PCFG: $P(S \rightarrow NP VP)=0.8$ = 80% of S rewrite as NP+VP; $P(PP \rightarrow P NP)=1.0$ =only rule for PP.

Attachment ambiguity: "saw the man with the binoculars" / "counts whales from space" — PP attaches to verb vs noun; parser may pick wrong head.

Information Extraction / NER (L5)

NER: PER/ORG/LOC. **IO** scheme fails when 2 same-type entities are adjacent ("Nancy Ken Follett" looks like 1). **IOB** fixes: **B-** starts each entity, **I-** continues.

CRF scores $\rightarrow$ probs via **softmax:** [1000, 1, 2, -30] $\rightarrow \approx$ [1, 0, 0, 0].

Label bias: locally-normalized models (MEMM) favor low-entropy states (few outgoing transitions) regardless of input; CRF avoids it via global normalization.

Encoders / BERT (L3 ★)

Foundation model: self-supervised pretrain, reuse backbone.

Static (word2vec) vs **contextual** (BERT) embeddings.

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

$\sqrt{d_k}$ = numerical stability. Multi-head: A heads $d_h = d/A$. BERT-Base $d=768$, $A=12$, $d_h=64$.

FFN $W_2\sigma(W_1x+b_1)+b_2$ (768 $\rightarrow$ 3072 $\rightarrow$ 768): processes **each token independently** with nonlinearity; residual $x+\text{Sub}(x)$; LayerNorm.

Q=info a token wants (query), **K**=what it offers, **V**=content passed on; attend strength = $q_i \cdot k_j$. Multi-head $\Rightarrow$ learn different relations. Encoder output = **one contextual vector per token**.

Params $\approx (V \times H) + 12LH^2$ (embeddings + L blocks).

BERT: encoder-only, bidirectional. MLM mask **15%**; NSP (weak, dropped later).

Fine-tune classif: prepend [CLS] $\rightarrow$ run $\rightarrow \hat{y} = \text{softmax}(Wh_{[\text{CLS}]}+b) \rightarrow \text{argmax}$.

Pooling: mean $\frac{1}{n} \sum h_i$, max.

SBERT (bi-encoder): encode sentences independently, $\cos(f(s_1), f(s_2))$, precomputable $\rightarrow$ fast. Fixes BERT $O(N^2)$ pair cost & anisotropy. Triplet loss $\max(0, d(a, p) - d(a, n) + m)$.

Cross-encoder=[CLS] s_1 [SEP] s_2 , accurate but slow (rerun each pair). Bi-encoder=fast cosine search.

Alignment: RLHF (L7 ★★)

Pipeline: prompts $\rightarrow$ **SFT** $\rightarrow$ human preferences $\rightarrow$ **reward model** $\rightarrow$ RL (**PPO**).

IFT/SFT: supervised on (prompt, good-resp); gives instruction-following + start policy. Self-Instruct = bootstrap data (label-first for classification).

Why beyond IFT: alignment is **comparative** (which is better?) $\rightarrow$ learn from preferences.

Reward model $r(x, y)$ scalar; can be smaller than LM; trained on pairwise comparisons $y_w \succ y_l | x$.

Bradley-Terry: $p^*(y_1 \succ y_2 | x) = \frac{e^{r(x, y_1)}}{e^{r(x, y_1)} + e^{r(x, y_2)}} = \sigma(r_1 - r_2)$, $\sigma(z) = \frac{1}{1+e^{-z}}$. Depends only on reward **difference**.

$$L_R = -\mathbb{E}_{(x, y_w, y_l)} [\log \sigma(r(x, y_w) - r(x, y_l))]$$

Describe the loss: score both responses, sigmoid of score-difference = P(winner beats loser); max-likelihood of observed human choices; small loss when winner scored $\gg$ loser, large when ranked wrong. Not regression/MSE: humans give relative not absolute scores.

Numbers: $r_w=2.0, r_l=1.2 \Rightarrow \sigma(0.8)=0.69$, loss $-\log 0.69=0.37$. Wrong way $\sigma(-0.5)=0.38$, loss 0.97.

PPO reward (KL-reg):

$r(x, y) = r_\phi(x, y) - \beta D_{KL}[\pi_\theta \| \pi_{SFT}]$
obj: $\max_\theta \mathbb{E}[r_\phi] - \beta D_{KL}[\pi_\theta \| \pi_{SFT}]$.

Role of β : balance reward vs staying near reference. High β =stay close (safe); low β =drift, risk **reward hacking**.

KL Divergence (★★ compute)

$D_{KL}(P \| Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$
 ≥ 0 ; $= 0$ iff $P=Q$; **not symmetric**. P =true/ground-truth, Q =model/measured. Use $\log_2$ for **bits** ($\log_2 z = \ln z / \ln 2$).

Ex: $P=[.5, .3, .2]$, $Q=[.4, .4, .2]$:
 $.5 \log_2 \frac{.5}{.4} + .3 \log_2 \frac{.3}{.4} + 0 = .1609 - .1245 = .036$ bits.

$Q=[.85, .10, .05]$: $-.383 + .476 + .4 = .493$ bits (worse).

Cross-entropy $H(p, q) = -\sum_x p(x) \log q(x) = H(p) + D_{KL}(p \| q)$. CE **loss** (one-hot true) $= -\log q_{\text{true}}$. Minimizing CE $\equiv$ minimizing KL to the data.

Hallucination types (L10)

Factuality = false vs the world. **Faithfulness** = inconsistent with prompt/context/own reasoning.

DPO — 13-step derivation (L8 ★★)

One sentence: DPO shifts probability mass toward completions humans like and away from disliked ones, *directly* from preference pairs—no reward model, no RL.

Why: RLHF has many parts (reward model + PPO loop + instability). DPO = simple classification-style loss, stable, cheap.

Steps:

1. Start KL-reg obj: $\max_\pi \mathbb{E}[r] - \beta D_{KL}[\pi \| \pi_{ref}]$.
2. Expand KL $\rightarrow \mathbb{E}[r - \beta \log \frac{\pi}{\pi_{ref}}]$.
3. To min: $\min_\pi \mathbb{E}[\log \frac{\pi}{\pi_{ref}} - \frac{1}{\beta} r]$.
4. Define **partition fn** $Z(x) = \sum_y \pi_{ref}(y|x) e^{r/\beta}$.
5. Target $\pi^* = \frac{1}{Z(x)} \pi_{ref} e^{r/\beta}$ (valid dist).
6. Obj $\rightarrow \min_\pi \mathbb{E}[D_{KL}(\pi \| \pi^*) - \log Z(x)]$.
7. $\log Z$ indep of $\pi \Rightarrow \min D_{KL}(\pi \| \pi^*)$; $KL \geq 0 \Rightarrow \pi = \pi^*$.
8. Invert: $r(x, y) = \beta \log \frac{\pi^*}{\pi_{ref}} + \beta \log Z(x)$.
9. Bradley-Terry: only reward **differences** matter.
10. $Z(x)$ **cancels** in the difference (same for y_w, y_l).
11. $p(y_w \succ y_l | x) = \sigma\left(\beta \log \frac{\pi^*(y_w)}{\pi_{ref}(y_w)} - \beta \log \frac{\pi^*(y_l)}{\pi_{ref}(y_l)}\right)$.
12. Parameterize $\pi^* \rightarrow \pi_\theta$.
13. DPO loss (below).

$$L_{DPO} = -\mathbb{E}\left[\log \sigma\left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)}\right)\right]$$

Closed-form optimal policy: $\pi^* = \frac{1}{Z(x)} \pi_{ref} e^{(1/\beta)r}$.

DPO — Puccinelli Q&A (memorize)

- **Why introduce $Z(x)$?** to **normalize** the reward-tilted π_{ref} so π^* sums to 1 (we shift mass $\Rightarrow$ must renormalize).
- **Why does it disappear?** BT uses reward **difference**; $\beta \log Z(x)$ depends only on x (same prompt) $\Rightarrow$ identical for $y_w, y_l \Rightarrow$ cancels.
- **Why need it gone?** $Z(x) = \sum$ over **all completions** = intractable.

- **Why "normalization constant"?** rescales weights to sum 1; constant in y but **changes with prompt x** .
- **Role of β ?** strength of reward tilt vs reference proximity (implicit KL).
- **Role of win term?** $\log \frac{\pi_\theta(y_w)}{\pi_{ref}(y_w)}$ raises preferred response prob vs ref; loser term lowers rejected.

Benchmarking (L10 ★)

Benchmark = dataset(s)+metric.
GLUE multi-task NLU.

Task→metric: CoLA→**MCC**;
 SST-2→acc; MRPC→acc/F1;
 STS-B→**Pearson/Spearman**;
 MNLI/QNLI/RTE/WNLI→acc.

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$$

Uses all 4 cells (F1 ignores TN). Ex: 90/10, always "yes": acc=.90 but **MCC=0**.

$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$, d_i =rank diff. Reversed ranks $\Rightarrow \rho = -1$.

NLI: premise+hyp $\rightarrow$ entail/contradict/neutral.
 RTE=binary. WNLI=Winograd pronoun.

Evolution: GLUE→SuperGLUE→HellaSwag→MMLU(57 subj)→MMM(U multimodal). **Contamination**: model saw test online $\rightarrow$ inflated. **LiveBench**: rolling, objective scoring.

Text-to-Speech (L14)

Two stages (trained separately):
 1) text→**mel-spectrogram** (Tacotron 2: char emb→conv→biLSTM→attn→AR decoder, 80-dim log-mel + stop).
 2) mel→waveform = **vocoder** (WaveNet: AR per-sample $P(x_t|x_{<t})$, dilated causal conv, μ -law 256 cls).

TTS speaker-dependent; ASR speaker-indep. Needs text normalization.

MOS: humans rate naturalness 1–5, MOS=mean. **A/B test**: same sentence, pick better, tally. No good automatic TTS metric.

mel $m = 2595 \log_{10}(1+f/700)$.

Audio basics: pitch $\approx \log(\text{freq Hz})$; loudness $\approx \log(\text{intensity})$, dB.

Spectrogram=|STFT| over time (x=time, y=freq). **MFCC**: mel filterbank $\rightarrow \log \rightarrow$ **DCT** = spectral envelope (perceptual).

Tokenization & Embeddings (L1)
 Subword balances OOV vs seq length.

BPE: merge **most frequent pair** iteratively. **WordPiece**: best score $\frac{P(xy)}{P(x)P(y)}$, “##”. **Unigram**: drop low-impact tokens. SentencePiece: raw text, space=“_”.

n-gram (m): $P(w_k|w_{k-m..k-1})$, MLE=counts; zeros→smoothing/backoff.

PP(W) = $P(w_1..w_N)^{-1/N} = 2^{H(W)}$
 lower=better; $\downarrow$ as order $\uparrow$.

word2vec: CBOW (predict center, fast) vs Skip-gram (predict context, more samples). SGNS neg-sample $\propto f^{0.75}$. fastText=char n-grams.

MT / Transformers (L2)

RNN $h_t = g(Uh_{t-1} + Wx_t)$; vanish/explode→LSTM gates (forget/input/output).

Seq2seq bottleneck = final encoder state. **Attention**: $\alpha = \text{softmax}(e)$, $\text{ctx} \sum \alpha_i h_i$.

Transformer: $Z = \text{softmax}(QK^\top / \sqrt{d_k})V$, multi-head (8, $d=512$, $d_k=64$);

masked decoder self-attn ($-\infty$ above diag); cross-attn (Q dec, K,V enc); **positional enc** (attn is permutation-invariant); FFN+residual+LN.

Train parallel (teacher forcing); infer autoregressive. KV-cache; GQA/MQA share K/V.

Trap: attn for token i uses query q_i + **all keys**.

Text Generation / Decoding (L6)

Greedy argmax; **beam** keep top- k paths (directed: MT/summary).

Temperature $P \propto e^{u_i/t}$; $t < 1$ sharpen, $t > 1$ flatten.

Top-k: among k highest (fixed).

Top-p: smallest set $\text{cum} \geq p$ (adaptive). Sampling=open-ended.

LoRA: $h = x(W + AB)$, freeze W , params $r(d+k) \ll dk$. QLoRA = +4-bit base.

Instruction-tune: CE on (instr,resp), mask instr tokens.

In-Context Learning (L9)

ICL: condition with examples, **no weight update**; emergent at scale. zero/one/few-shot.

Roles: System (rules), User (task/data), Assistant (outputs).

CoT “think step by step” $\rightarrow$ better math/logic. **Self-consistency**=sample $\times$ N + majority vote.

ReAct=Thought→Action→Observation. Tips: simple prompts, instructions over many constraints, document (model,temp,top-p).

RAG (L11)
 Retrieve chunks $\rightarrow$ augment prompt $\rightarrow$ generate. Fixes stale/parametric knowledge w/o retraining.

Pipeline: chunk→embed→index(vector DB)→embed query→similarity→prompt.

Bi-encoder/SBERT retrieve (fast, precompute); **cross-encoder** rerank (accurate, slow).

$\cos(x, y) = \frac{x^\top y}{\|x\| \|y\|}$; normalized $\Rightarrow \cos \equiv \text{dot} \equiv \text{rank}$; $\|x - y\|^2 = 2 - 2 \cos$.

Sparse: **BM25**. Hybrid: **RRF** $\sum_i \frac{1}{k + r_i(d)}$. ANN: IVF (k-means), HNSW (graph).

MRR = $\frac{1}{|Q|} \sum \frac{1}{\text{rank}_i}$; **NDCG** = $\frac{DCG}{IDCG}$, $DCG = \sum \frac{2^{\text{rel}_i - 1}}{\log_2(i+1)}$.

RAGAS: Faithfulness = (claims supported)/(claims in answer).

Agentic AI (L12)
 Agent: sets goals, plans, **takes actions** (vs chatbot responds). Memory: short=context, long=vector DB/RAG. Tools: search/code/APIs.

ReAct loop. Evaluate **full transcript** (not just final).

pass@k=success over k trials. Risks: cascading failures, over-privilege, reward hacking.

Speech-to-Text / ASR (L13)
 audio→log-mel (~ 80 ch, 25ms). $dB = 10 \log_{10}(I/I_{min})$.

wav2vec2: CNN+Transformer, self-supervised contrastive + product quantization, fine-tune **CTC**.

CTC: blank token; many alignments→one output (collapse repeats, drop blanks); loss

– $\log \sum_{align} \prod_t p(a_t|h_t)$; add LM at .667.
decode.

Whisper: enc-dec, weak supervision (680k h), multitask; robust, can loop.

$$\text{WER} = \frac{S+D+I}{N}.$$

Numeric quick-reference (calc)

Sigmoid $\sigma(z) = \frac{1}{1+e^{-z}}$:
 $\sigma(0)=.50$, $\sigma(.5)=.62$, $\sigma(.8)=.69$,
 $\sigma(1)=.73$, $\sigma(1.5)=.82$, $\sigma(2)=.88$.
 $\sigma(-z)=1-\sigma(z)$.

– $\log(\cdot)$ (NLL): – $\log .69=.37$,
– $\log .5=.69$, – $\log .38=.97$,
– $\log .9=.105$.

log₂: $\log_2 1.25=.322$, $\log_2 .75=$ –
.415, $\log_2 1.5=.585$, $\log_2 2=1$,
 $\log_2 3=1.585$, $\log_2 4=2$. ($\log_2 z =$
 $\ln z / \ln 2$, $\ln 2=.693$.)

Worked: classification

5 sentences, classes CH/FR/DE.
GT→Pred: CH→CH, FR→DE,
DE→CH, CH→FR, FR→FR.

CH: TP1,FP1,FN1⇒ $P=R=\frac{1}{2}$, $F1=\frac{1}{2}$.
FR: same $F1=\frac{1}{2}$. DE: TP0⇒ $F1=0$.

Supports CH2,FR2,DE1. macro=
 $\frac{.5+.5+0}{3} = \frac{1}{3}$. weighted=
 $\frac{2(.5)+2(.5)+0}{5} = .4$.

Worked: UAS/LAS

6 tokens, (gold head,label | pred):
t4 right head wrong label, t5 wrong
head; rest correct.

Correct heads = 5/6 ⇒ **UAS** = .833.

Correct head+label = 4/6 ⇒ **LAS** =

BLEU (L1/L2)

$\text{BLEU} = BP \cdot \exp(\sum_n w_n \log p_n)$,
 $BP = \min(1, e^{1-r/c})$, $p_n=\text{clipped}$
n-gram precision (cap match count
by ref count). Precision-based;
ROUGE=recall-based.

Worked: RLHF / MCC / Spear- man

Reward NLL: $r_w=2.0, r_l=1.2$:
 $P=\sigma(0.8)=.69$, loss $-\log .69=.37$
(good). $r_w=1.0, r_l=1.5$: $\sigma(-.5)=.38$,
loss .97 (bad).

MCC: 90 ok/10 not, always "ok":
TP90,FP10,TN0,FN0; acc= .90 but
 $\text{MCC} = \frac{90 \cdot 0 - 10 \cdot 0}{\dots} = 0$.

Spearman: $R = [1, 2, 3, 4, 5]$,
 $S = [5, 4, 3, 2, 1]$: $d^2 =$
 $[16, 4, 0, 4, 16], \sum = 40$; $\rho = 1 - \frac{6 \cdot 40}{5 \cdot 24} = -$
1.

Worked: retrieval / WER

NDCG@5, $\text{rel} = [0, 3, 2, 1, 0]$:
 $\text{DCG} = 0 + \frac{2^3-1}{\log_2 3} + \frac{2^2-1}{\log_2 4} + \frac{1}{\log_2 5} + 0 =$
 $0 + 4.42 + 1.5 + 0.43 = 6.35$;
 $\text{IDCG} = 9.39 \Rightarrow \text{NDCG} = .68$.

WER: ref 10 words, 1S+1D+2I
⇒ $(1+1+2)/10 = .40$.

Cosine: unit vecs, $\cos=.6 \Rightarrow$
 $\|x-y\|^2=2-1.2=.8$.

LoRA: $4096^2=16.7\text{M}$ full; $r=4 \Rightarrow$
 $4(4096+4096)=32768$ ($\sim 512 \times$
fewer).

Fact bank / traps

- Lower perplexity = better LM (↓ with n-gram order).
- BPE merges **frequent pair**; WordPiece by **score**; Unigram **prunes**.
- Temp $t<1$ sharpen, $t>1$ random; top- k fixed, top- p adaptive.
- Attn weights: **query q_i vs all keys**; $\sqrt{d_k}$ =stability.
- BLEU=precision+BP; ROUGE=recall+precision.
- KL: not symmetric, ≥ 0 , = 0 iff equal; use **bits**.
- $\text{LAS} \leq \text{UAS}$ always.
- Accuracy misleads on imbalance / missing classes (use macro-F1, MCC).
- DPO: no reward model, no RL; partition fn cancels.
- Bi-encoder=retrieve(fast); cross-encoder=rerank(accurate,slow).
- RLHF reward = $r_\phi - \beta \text{KL}$; high β =stay near SFT.

Exam tactics

Need only **3.5/6** (5.5 pregrade @25%). Bank MC + quick calcs first, then text answers. State assumptions if unclear. KL in **bits** ($\log_2$). Show formula+plug-in. **Must-compute**: P/R/F1, macro/weighted, UAS/LAS, KL(bits), reward/DPO sigmoid-NLL, MCC, Spearman, WER, perplexity, cosine, NDCG.